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System and Method for Predictive Supply Chain Management

Abstract

Various embodiments of the present invention comprise a supply chain platform configured in connection with an inventory hub, a supplier system, an enterprise data platform, and a customer system to respond to disturbance events within a supply chain. Such a supply chain platform may collect a variety of supply chain data to determine a risk score relating to such disturbance event(s) and one or more mitigation opportunities in response thereto via a mitigation module configured to generate one or more scenarios and assess the same for efficacy and readiness. Such mitigation opportunities may include the use of one or more supplier sites within the supply chain. Such a supply chain platform may be configured to generate a plurality of reports having interactive data visualizations configurable according to key element data and/or parameter input(s) supplied by a user through a user device and may differentiate between important and inconsequential disturbance events.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present Non-Provisional application claims priority to the earlier-filed and currently Provisional Application No. 63/562,587 having a filing date of Mar. 7, 2024, the contents of which are hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Supply chain management, namely the management of the flow of goods and services which transform raw materials into finished products for consumers, is equally as important and it is complex. Failure to adequately manage a supply chain may result in significant delays, increase operating and/or production costs, and potentially jeopardize client relationships. And while the complexity of supply chains may derive from a business itself, complexity often arises from unforeseen and uncontrollable external factors. Indeed, business-related disruptions—e.g., factory disruptions, labor disruptions, sales disruptions, supplier disruptions, mergers and acquisitions, and legal actions—are only one piece of the puzzle as pandemics, wars, famines, and natural disasters similarly serve to disrupt supply chains. Any one of the foregoing disruptions may quickly and easily work its way through a supply chain as a delay in one supply chain node may impact the timeliness of work at another node. Consequently, adequate management of a supply chain is imperative for many, if not all businesses. And adequate management of a supply chain mandates quick, decisive, and accurate decision making in response to the myriad threats which may negatively effect production and delivery.

[0003] For certain organizations, the complexity of supply chain management increases as a result of the size and nature of the business. Efforts to reduce complexity, and thereby improve response time, production time, and resource efficiency, for many large organizations often involves supply chain integration through which an organization attempts to bring various links within a supply chain closer together. In so doing, integration aims to increase the sharing of information between the supply chain links and/or nodes, more efficiently allocate costs, and reduce response times to any issues that may arise.

[0004] Vertical integration is one such way a company may seek to better integrate the various parts of its business. Through vertical integration, a company seeks to control multiple stages of its production process, whether by taking greater control over its suppliers and/or by taking greater control over its distributors. While vertical integration may be effective at reducing costs, improving efficiency, and offering a company greater control over its supply chain, such vertically integrated organizations often run into difficulties with supply chain management. For instance, vertical integration requires much upfront investment and thus increases risk. Further, because vertically integrated organizations control larger portions if not the entirety of their supply chain, such organizations are often cumbersome and inflexible. As a result, vertically integrated organizations often respond slowly to changes, both in the market and to disruptions in their supply chain. Without third-party suppliers and/or distributors to easily turn to, any such issues can often infect the entirety of the business in the absence of adequate planning.

[0005] Such difficulties are often exacerbated for organizations having disparate business groups utilizing disparate technological solutions, or where the flow of information is often stilted. For instance, different businesses within the organization may use different enterprise resource planning systems, such as when the organization services a vast and varied customer base mandating different approaches to resource planning. Similarly, disjointed communication channels and obstructive approval processes may stem the flow of information and disrupt prompt decision making. As a result, such vertically integrated organizations often produce labored responses in the face of a disruptive event, which naturally wastes resources, whether by producing unnecessary stock of a given product or failing to timely source a replacement component according to the needs of its customers. Hence, forecasting risk and enabling swift, responsive action is imperative to avoid delays and disruptions in the supply chains of vertically integrated organizations.

[0006] Conventional supply chain management systems aim to resolve some of these issues via forecasting tools and real-time data management to facilitate decision making. Such conventional systems typically focus on logistics, purchasing, and enterprise resource planning, with the goal of automating ways to manage a supply chain network. However, such conventional systems are often insufficient for the type of vertically integrated organizations discussed heretofore, as their one-size-fits-all approach fails to account for the unique needs of such organizations. For instance, such conventional systems often fail to account for issues relating to the flow of information and fail to appreciate the inflexible nature of such organizations. Similarly, such conventional systems commonly over-signal potential issues, lest they miss something important. Yet, this inundation of data often means key issues are overlooked or dismissed, whereas nominal issues are readily resolved. The end result is a system often lacking sufficient insight and solutions, and one which distracts users from the important issues.

[0007] In view thereof, there exists a need in the art for a supply chain management system configured to increase the efficiency and adaptability of supply chains operated by vertically integrated organizations. Such a system should be configured to provide greater visibility across an organization, provide predictive analyses, and develop prescriptive plans in response to a disruptive event. Furthermore, such a system should be configured to provide insights between various segments of the organization and harmonize designs and strategies to facilitate site-to-site sharing. Such a system should further be user-configurable to enable specific parameterization of the analyses of outputs thereof. Additionally, such a system should be configured to quickly and accurately identify the important from the mundane, thereby enabling users to quickly address key issues.

BRIEF SUMMARY

[0008] Various embodiments of the present invention are directed to a supply chain management system configured to resolve one or more of the aforementioned needs in the art, such as increasing visibility and adaptability in response to supply chain disturbances across vertically integrated supply chains and differentiating between important and inconsequential supply chain disruption events.

[0009] For instance, in at least one embodiment of the present invention, such a supply chain management system may comprise: an inventory management hub communicatively configured in connection with a supply chain platform; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub; an analytics system configured to determine at least one risk score according to a risk calculation routine performed on said supply chain data; said analytics system further configured to generate at least one mitigation opportunity according to a mitigation module; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity.

[0010] In at least one alternative embodiment of the present invention, such a supply chain management system may comprise: an inventory management hub communicatively configured in connection with a supply chain platform; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub; a historical data module configured to generate historical data; an analytics system comprising a risk and recovery module configured to determine at least one risk score according to a risk calculation routine performed on said supply chain data and said historical data; said analytics system further comprising a mitigation module configured to generate at least one mitigation opportunity according to said supply chain data and said historical data; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity.

[0011] In yet additional embodiments of the present invention, such a supply chain management system may comprise: an inventory management hub communicatively configured in connection with a supply chain platform and a supplier system, said supplier system comprising a primary supplier site and at least one secondary supplier site, said primary supplier site having constraint data indicating a potential disturbance; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub and said constraint data from said primary supplier site; a historical data module configured to generate historical data; a dashboard module comprising a key element component configured to receive key element data from said at least one user device and a parameter component configured to receive at least one parameter input from said at least one user device; an analytics system comprising a risk and recovery module configured to determine, in response to said constraint data, at least one risk score and at least one risk classification according to a risk calculation routine performed on said supply chain data and said historical data; said analytics system further comprising a mitigation module configured to generate at least one mitigation opportunity, said at least one mitigation opportunity generated according to a scenario module configured to determine at least one scenario dataset and a scenario efficacy module configured to determine an efficacy metric and a deployment metric for said at least one scenario dataset; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity, and configurable according to said at least one parameter input. As may be understood, the foregoing embodiments are merely exemplary.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0012] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0013] FIG. **1**A illustrates a block diagram of a supply chain platform in accordance with at least one embodiment of the present invention.

[0014] FIG. **1**B illustrates a block diagram of a communicative configuration between a supplier system, an inventory management database, an enterprise data platform, a customer system, an administrator system, and a supply chain platform, in accordance with at least one embodiment of

the present invention, such as the one depicted in FIG. **1**A.
[0015] FIG. **1**C illustrates a block diagram of a user device which may be used in connection with the various systems, databases, and platforms shown in the embodiments depicted in FIG. **1**A and FIG. **1**B.
[0016] FIG. **2** illustrates a block diagram of the supply chain of an organization, in accordance with at least one embodiment of the present invention.
[0017] FIG. **3**A illustrates a block diagram of a supply chain platform, in accordance with at least one embodiment of the present invention.
[0018] FIG. **3**B illustrates a block diagram of a data receipt module, in accordance with the embodiment depicted in FIG. **3**A.
[0019] FIG. **3**C illustrates a block diagram of an analytics system, and the risk and recovery module and inventory entitlement module thereof, in accordance with the embodiment depicted in FIG. **3**A.
[0020] FIG. **3**D illustrates a block diagram of one embodiment of a mitigation module, in accordance with the embodiments depicted in FIGS. **3**A-**3**C.
[0021] FIG. **3**E illustrates a block diagram of one alternative embodiment of a mitigation module, in accordance with the embodiments depicted in FIGS. **3**A-**3**C.
[0022] FIG. **4**A illustrates an output of the glidepath module, in accordance with at least one embodiment of the present invention.
[0023] FIG. **4**B illustrates an output of the material risk module, in accordance, in accordance with at least one embodiment of the present invention.
[0024] FIG. **4**C illustrates a user interface representation of an aspect of the subject matter in accordance with one embodiment.
[0025] FIG. **4**D illustrates an output of an interactive data visualization, in accordance with at least one embodiment of the present invention.
[0026] FIG. **4**E illustrates an output of a material risk module, in accordance with at least one embodiment of the present invention.
[0027] FIG. **4**F illustrates an output of a shipping status module, in accordance with at least one embodiment of the present invention.
[0028] FIG. **4**G illustrates an output of a quarterly status module, in accordance with at least one embodiment of the present invention.
[0029] FIG. **4**H illustrates an output of a surplus status module, in accordance with at least one embodiment of the present invention.
[0030] FIG. **4**I illustrates an output of an expiration status module, in accordance with at least one embodiment of the present invention.
[0031] FIG. **4**J illustrates an output of a portfolio module, in accordance with at least one embodiment of the present invention.
[0032] FIG. **4**K illustrates an output of a portfolio module, in accordance with at least one embodiment of the present invention.

DETAILED DESCRIPTION

[0033] Various embodiments of the present disclosure are directed to a supply chain system **100** generally configured to provide end-to-end response and mitigation strategies to supply chain disturbances, such as those resulting in material constraints, as well as provide numerous reports and tools through which a user may both investigate and tailor a supply chain response to the particular situation at-hand. In at least some embodiments, the supply chain system **100** of the present disclosure may include a supply chain platform **302** configured to determine a risk associated with a disturbance event, appropriately tailor inventory entitlement to limit excess inventory, and utilize scenarios to develop and suggest mitigation strategies to such disturbance event. Even further, various embodiments of such a supply chain platform **302** may include a variety of reports and other interactive data visualization(s) **404** through which a user may identify risks and the components thereof, how these risks may impact a customer, and how a given mitigation strategy may affect both the customer at issue, as well as the organization at large. And finally, various embodiments of such a supply chain platform **302** may be configured to identify, signal, or otherwise differentiate between key events and exceptions—i.e., those events offering only minimal threat of a supply chain distribution.

[0034] In the following detailed description, reference is made to the accompanying drawings that form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made, without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

[0035] The description may use phrases such as “an embodiment,” “various embodiments,” and “some embodiments,” each of which may refer to one or more of the same or different embodiments. Furthermore, the terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. When used to describe a range of values, the phrase “between X and Y” represents a range that includes X and Y. As used herein, an “apparatus” may refer to any individual device, collection of devices, part of a device, or collections of parts of devices. The drawings are not necessarily to scale.

[0036] Depicted in FIG. **1**A is an exemplary embodiment of a supply chain system **100** in accordance with the present disclosure. Specifically, the supply chain system **100** of various embodiment of the present disclosure may include a supply chain platform **302** communicatively configured in connection with a variety of systems, databases, and platforms for the collection and transmission of relevant data through at least one network **112**, such as the internet, an intranet and/or wireless network, a wireless local area network, a metropolitan area network, and/or some other wireless network akin thereto. Further, it may be understood each such system and/or platform may be operable by a user(s) through user device(s) **114** configured to communicate over such network **112**.

[0037] For instance, such a supply chain platform **302** may be communicatively coupled with: (a) an inventory management database **102**, which may be configured to store, at least in part, inventory-related data from a plurality of manufacturing sites, shipping sites, and other related nodes along a supply chain; (b) an enterprise data platform **106**, which may comprise one or more systems storing and/or transmitting data relating to the organization owning, running, presiding over, or otherwise operating the supply chain platform **302**; (c) a supplier system **104**, which may comprise the various systems of one or more entities configured to supply the resulting supply chain with raw materials, products, or other goods related to a purchase order from a customer; (d) a customer system **108**, which may comprise one or more systems configured to store and/or transmit data relating to a given customer, such as the party placing a purchase order with the organization; and (e) an administrator system **110**, which may comprise a system configured to store and/or transmit data relating to an administrator of the supply chain platform **302**, such as an employee and/or individual of the organization operating the supply chain platform **302**. As may be understood, the foregoing systems and/or platforms may be distributed over multiple databases and/or systems, including one or more servers. Each of these systems, platforms, and databases will be discussed in greater detail hereafter.

[0038] With continued reference to FIG. **1**A and further reference to FIG. **1**B, it may be seen at least one embodiment of the present disclosure may be configured such that certain systems, databases, and platforms are communicatively isolated and/or distinct from each other. For instance, in the embodiment depicted in FIG. **1**B, the supplier system **104** and enterprise data platform **106** are each solely communicatively coupled with the inventory management database **102**, which may thus be configured to collect and aggregate the relevant data flowing therefrom. Likewise, the customer system **108** and the administrator system **110** may each be solely communicatively coupled with the supply chain platform **302**. As a result, each such entity—e.g., customer, supplier, and administrator—may communicate with the supply chain platform **302** while maintaining at least some degree of blinding and/or confidentiality as to the other's data. Finally, the inventory management database **102** and the supply chain platform **302** may be communicatively coupled together, thereby enabling the supply chain platform **302** full access to the relevant data from the supplier system **104** and enterprise data platform **106**. As may be understood, alternative arrangements of communicative coupling between the various systems, databases, platforms, and/or user device(s) **114** of the present disclosure are envisioned herein.

[0039] In at least one embodiment, such as that depicted in FIG. **1**A, such a supply chain platform **302** may generally comprise a supply chain server system **128**, which may itself comprise some computer-related structure and/or entity configured to perform one or more specific functionalities, whether based in hardware, software, or some combination thereof. For example, such a supply chain platform **302** may be configured as a process running on a processor, a processor itself, a hard disk drive, a plurality of storage drives, an object, an executable, a computer-executable program, and/or a computer. In at least one embodiment, such a supply chain platform **302** may be configured to operate using and/or in connection with a relational database management system and/or a cloud database such as Microsoft Azure SQL. Alternatively, such a supply chain platform **302** may instead be configured to operate using and/or in connection with the enterprise data platform **106** itself, such as via backend development configured thereon. Likewise, such a supply chain platform **302** may itself generally comprise a platform-as-a-service (Paas) and/or an application-platform-as-a-service (aPaaS). As may be understood, alternative such modes of configuration for such supply chain server system **128** are envisioned herein.

[0040] Likewise, the various systems and/or platforms communicatively configured in connection with the supply chain platform **302** may similarly comprise some computer-related entity configured to perform one or more specific functionalities, whether based in hardware, software, or some combination thereof. For example, the enterprise data platform **106** may be configured to operate using one or more cloud-based computing platforms, such as Amazon Web Services. The various systems of the supplier system **104**, customer system **108**, and/or administrator system **110**, meanwhile may each function according to their own computing preferences. Accordingly, it may be understood such systems and platforms should be construed as non-limiting according to their computing platforms and structures, as may be understood by those skilled in the art.

[0041] FIG. **1**C depicts at least one embodiment of a user device **114**, through which a user may interact with the supply chain platform **302** and/or any other interconnected systems and/or platforms. Such a user device **114** may comprise, for instance, a smart phone, laptop, computer, tablet, or other similar device, and may include a plurality of electrical and electronic components configured to provide power, operational control, and/or protection to the same. For instance, such a user device **114** may comprise a processing component **116**, such as an electronic microprocessor, microcontroller, or other similar device, a power component **118**, such as a battery, and a storage component **120**, such as a memory, including, without limitation, a non-transitory, computer-readable memory. Likewise, such a user device **114** may additionally comprise a display component **122**, such as a screen or touchscreen, and an interface component **124**, which may be configured to receive data and provide data to the user, such as in audible, textual, and/or graphical form. Finally, such a user device **114** may additionally comprise I/O devices **126**, such as a touchscreen, keypad, keyboard, cursor-controlled device, or other similar component configured to provide input-output functionality.

[0042] FIG. **2** depicts at least one embodiment a logistical structure **200**—i.e., a supply chain and/or organizational structure—for a given organization. In at least one embodiment, such a logistical structure **200** may be owned and/or operated by, for instance, the entity owning, running, and/or operating the supply chain platform **302**. Shown in the embodiment depicted in FIG. **2** is an exemplary logistical structure **200** for an organization having a vertically-integrated structure—i.e., a logistical structure **200** wherein the organization controls multiple stages of the production process and, thereby the supply chain itself, as discussed in greater detail heretofore. As may be understood, the depicted embodiment is merely exemplary and simplified for the sake of brevity. Notwithstanding, it may be understood constraint data **353** emanating from a primary supplier site **204** may flow through an inventory hub **202** having an inventory management database **102** operating therefrom and ultimately affect at least one distribution spoke(s) **206***a*-**206***n*. Such distribution spokes **206***a*-**206***n* may comprise one or more distribution centers and/or other entities, locations, and/or facilities associated with the logistical structure **200** of the organization, whether or not ostensibly configured to fulfill a particular order from a customer in relation to the primary supplier site **204**. As may be understood, the logistical structure **200** of various embodiments contemplated herein may include a plurality of primary supplier sites **204**, each interconnected with a number of distribution spokes **206***a*-**206***n* and, as discussed hereafter, secondary supplier sites **208***a*-**208***n*.

[0043] In view of the foregoing, it may be understood such constraint data **353** may comprise some data relating to a constraint which may and/or will affect the supply chain of an organization, such as an insufficient amount of raw material to generate a part. As may be understood, such constraint data **353** may result from any given supply chain disturbance, whether external in nature—e.g., war, famine, pandemic, regulatory, etc.—or internal in nature—e.g., personnel, site defects, breakage in manufacturing line, etc. Hence, it may be understood such constraint data **353** may include all information pertinent to the relevant complaint, such as, without limitation, the part number at issue, the projected deficit, the expected downtime, the customer order(s) impacted, etc.

[0044] With continued reference to FIG. **2**, it may be seen such an inventory hub **202** may further be interconnected with one or more secondary supplier sites **208***a*-**208***n*. Each such secondary supplier site **208***a*-**208***n* may comprise a location, such as a manufacturing location or a distribution center, which may be configured to respond to the constraint data **353** restricting the operations of the primary supplier site **204**. As may be noted, each such secondary supplier site **208***a*-**208***n* may comprise its own enterprise resource platform **212***a*-**212***n*, which may, in at least some embodiments, be distinct from any similar system and/or platform utilized by or otherwise communicating with the inventory management database **102** of the inventory hub **202**. With reference to FIG. **1**B, it should be understood such primary supplier site **204**, distribution spokes **206***a*-**206***n*, and secondary supplier site **208***a*-**208***n* may, in at least one embodiment, collectively comprise the supplier system **104** depicted therein.

[0045] Hence, it may be understood the supply chain platform **302** of various embodiments of the present disclosure may be configured to communicate with the inventory management database **102** when a primary supplier site **204** is experiencing a supply chain disruption event sufficient to generate constraint data **353**. In so doing, the supply chain platform **302** may gather all relevant information, whether from the administrator system **110**, the customer system **108**, or otherwise, to identify and implement a response to such disruption event, such as the use of at least one secondary supplier site **208***a*-**208***n* to assist the primary supplier site **204**.

[0046] In view thereof, it may be understood the supply chain platform **302** may be configured to collect and transmit a variety of data, as well as perform a variety of functions relating to the logistical structure **200** of the organization. In at least one embodiment of the present invention, such as the one depicted in FIG. **3**A, such a supply chain platform **302** may comprise certain modules and/or systems, such as: (a) a data receipt module **304**; (b) a dashboard module **306**; (c) a historical data module **308**; (d) an analytics system **310**; (e) a report module **312**; and (f) a security module **314**. Each of these modules and/or systems will be discussed in greater detail hereafter.

[0047] In at least one embodiment, such a data receipt module **304** may be configured to receive and store various portions of data from the various systems interconnected with the supply chain platform **302**, such as the customer system **108**, the administrator system **110**, and the inventory management database **102** (and thus the supplier system **104** and the enterprise data platform **106** as well). For instance, as depicted in FIG. 3B, at least one embodiment of such a data receipt module **304** may be configured to collect a variety of data from the aforementioned systems and/or platforms, all of which may be collectively referred to herein as supply chain data. By way of example, sales order data **341**, sales forecast data **343**, purchase order data **345**, work order data **346**, EDP data **347** (electronic data processing data), dependent demand data **349**, and MRP data **350** (material requirements planning data) may be collected from the enterprise data platform **106**, inventory hub **202**, and/or supplier system **104**. Likewise, inventory data **344**, pick list data **342**, and bill of materials data **348** may be collected from the supplier system **104** and/or inventory hub **202**. Historical data **351**, meanwhile, may be collected from the supply chain platform **302**, and more particularly from the historical data module **308** thereof, as will be discussed in greater detail hereafter. Likewise, key element data **352** may be collected from the supply chain platform **302** after being supplied from the customer system **108** and/or administrator system **110**. Moreover, in at least one embodiment, such key element data **352** may be automatically provided by the supply chain platform **302** according to a rules module, which may be configured to, for instance, automatically determine the key element data **352** according to at least one regulatory metric, such as any applicable laws, rules, and/or regulations. As may be understood, the foregoing types of data collected by the data receipt module **304** is merely exemplary, and various data may be collected from each of the systems and/or platforms interconnected with the supply chain platform **302** for their collection and storage within the data receipt module **304**.

[0048] As may be understood, due to the collection of a variety of data from a variety of sources, it is possible the data receipt module **304** of at least one embodiment of the supply chain platform **302** may receive data in a variety of formats, thereby leading to unstructured, redundant, or otherwise unclean data. Accordingly, the data receipt module **304** may, in at least one embodiment, comprise a data normalization routine through which the data receipt module **304** may revise and/or reorganize the data it receives to achieve a standardized data format.

[0049] Returning to FIG. 3A, at least one embodiment of the supply chain platform **302** of the present invention may comprise a dashboard module **306**. Such a dashboard module **306** may be configured in input/output communication with the user device(s) **114** operated by the administrator(s) of the administrator system **110**, the customer(s) of the customer system **108**, and/or the other general users of the supply chain platform **302**. For instance, such a dashboard module **306** may comprise a UI component **320** configured to generate one or more user interfaces to be displayed on the user device **114** and, more particularly, the display component **122** and/or interface component **124** thereof. Such UI component **320**, and the various user interfaces generated and displayed thereby, will be discussed in greater detail hereafter.

[0050] Such a dashboard module **306** may further comprise a key element component **316** and a parameter component **318**, each of which may be configured to enable a user—e.g., an administrator, a customer, or otherwise—to specifically interact with the supply chain platform **302**. For instance, the key element component **316** may be configured to enable a user to input key element data **352** into the supply chain platform **302**, which may enable such user to specifically identify which elements of the supply chain, and/or the products, goods, and materials flowing therethrough, are most impactful to the supply chain and/or its ability to fulfill any given purchase order. Conversely, such a parameter component **318** may be configured to enable a user to apply various parameters to the information presented by the supply chain platform **302**. For instance, such a parameter component **318** may allow the user to apply parameter data **354** to the relevant information via at least one parameter input. Such parameter data **354** may include, without limitation, changes to the applicable date range, revenue basis, allowable lapse periods, risk hedging, and exemptions to types of risk mitigation strategies. Such key element component **316** and parameter component **318** will be discussed in greater detail hereafter.

[0051] With continued reference to FIG. 3A, it may be seen at least one embodiment of the supply chain platform **302** of the present invention may additionally comprise a historical data module

308. Such a historical data module **308** may be configured to generate historical data **351** which may be used by the supply chain platform **302** in its analysis of disruption events, generation of mitigation opportunities, and display of various portions of information. Such historical data **351** generated by the historical data module **308** may comprise, for instance, the performance history of the primary supplier site **204**, which may include how the same responded to other disruptive events and whether the supply chain itself experienced a delay as a result thereof. As such, it may be understood such historical data **351** may generally include how a primary supplier site **204** has previously interacted with the various secondary supplier site(s) **208***a*-**208***n* and/or distribution spoke(s) **206***a*-**206***n*. Notwithstanding, alternative types, categories, and uses of historical data **351** are envisioned herein, such as any and all data and types thereof collected by the data receipt module **304**.

[0052] At least one embodiment of the supply chain platform **302** of the present invention, such as the one depicted in FIG. **3**A, may additionally comprise an analytics system **310**. Such an analytics system **310** may be configured to generally ingest the data collected by the data receipt module **304** and analyze the same to determine a variety of relevant outputs. For instance, such an analytics system **310** may determine response output(s) **398** including, without limitation, risk and recovery to future demand, prioritization to demand, inventory entitlement, risk mitigation opportunities, excess and obsolete predictions and actions relating thereto, glidepaths, and quarterly status updates. As may be understood, the foregoing response outputs **398** are merely exemplary, and additional outputs are contemplated herein.

[0053] In at least one embodiment, such an analytics system **310** may comprise a risk and recovery module **322** and an inventory entitlement module **324**, such as in the embodiment depicted in FIG. **3**C. Such a risk and recovery module **322** may be configured to identify potential risks to a supply chain and assess the impact of such a risk. For instance, such a risk and recovery module **322** may comprise a risk calculation routine **358** which may, upon receiving data related to a potential disturbance to a supply chain, perform an analysis to generate a risk score **360**, which itself may provide a metric indicating the risk of a delay somewhere along the supply chain. As may be understood, such a risk score **360** may, in at least one embodiment, consider the risks for specific distribution spokes **206***a*-**206***n*, as well as for the primary supplier site **204** itself. For instance, with consideration of the supply chain of the logistical structure **200** depicted in FIG. **2**, the constraint data **353** may flow into the risk and recovery module **322** and, in accordance with the data received by the data receipt module **304**, the risk calculation routine **358** may generate a risk score **360** indicating whether any sales order data **341** residing at one or more of the distribution spokes **206***a*-**206***n* is at risk of delay dependent, at least in part, on the historical data **351** relating to the primary supplier site **204**. Such a risk calculation routine **358** may utilize a variety of indicators based on the data residing within the data receipt module **304** to determine the risk score **360**, such as the inventory data **344**, sales forecast data **343**, and dependent demand data **349** indicating whether this particular disturbance may effect later sales, or bill of materials data **348**, to assess whether this disturbance may impact other products contemplated in sales order data **341**, and pick list data **342** to assess whether alternative options exist for collecting items to fulfill the relevant orders. As may be understood, the above operations of the risk calculation routine **358** is merely illustrative, and alternative operations are contemplated herein.

[0054] In conjunction therewith, at least one embodiment of such a risk calculation routine **358** may further comprise a risk classification **362**, which may provide some classification for the risk score **360** calculated thereby. For instance, such a risk classification **362** may be used to indicate whether a risk is manageable, whether it should be monitored and/or mitigated, or whether it requires immediate action. In at least one embodiment, such a risk classification **362** may be individually tailored to a given distribution spoke **206***a*-**206***n*, or to a particular user, such as a customer within the customer system **108**. Accordingly, such a risk classification **362** may consider any key element data **352** and/or parameter data **354** into the supply chain platform **302** by a user through the dashboard module **306** or otherwise generated by the supply chain platform **302**.

[0055] Further, such a risk and recovery module **322** may additionally comprise a demand prioritization routine **356** which may be configured to determine which demands and/or orders should be prioritized in the face of a given disturbance. Accordingly, such a demand prioritization routine **356** may utilize various data to determine a demand priority metric indicating whether sales order data **341** should be fulfilled or accepted, or whether a later fulfillment is preferable, amongst other similar prioritization tactics. As with the risk classification **362**, such a demand prioritization routine **356** may consider key element data **352** and/or parameter data **354** input into and/or generated by the supply chain platform **302**, such as the user's intention to avoid any delays greater than a set period of time, in making its demand prioritization decisions.

[0056] With continued reference to FIG. **3**C, it may be seen the at least one embodiment of such an inventory entitlement module **324** of the analytics system **310** may comprise an inventory quantity routine **364**, which may be configured to determine an appropriate minimum amount of stock for the supplier system **104** to have on hand. In at least one embodiment of the present invention, such an inventory quantity routine **364** may comprise an inventory classification score **366**, which may be used to determine the importance of a particular type of inventory. For example, an item found in the bill of materials data **348** of numerous products may be deemed more important than an item used in connection with only one product. Likewise, an item used in connection with only one product with a larger amount of sales order data **341** may be deemed more important than an item used in connection with many products each having sales forecast data **343** indicating minimal purchases over the next period of time. Further, such an inventory quantity routine **364** may additionally calculate a service level score **368**, which may be configured to determine the importance a user—e.g., an administrator, customer, and/or supplier—may place on a given product, location, or customer. For instance, a user may input key element data **352** into the dashboard module **306** indicating more or less emphasis to be placed on on-time delivery, which may be used to adjust the minimum quantity of inventory the primary supplier site **204** and/or secondary supplier sites **208***a*-**208***n* should keep in stock.

[0057] Various embodiments of such an analytics system **310** of the supply chain platform **302** may additionally comprise a mitigation module **326**. Such a mitigation module **326** may be configured to determine one or mitigation opportunities in view of a disturbance event. For instance, as depicted in FIG. **3**D, such a mitigation module **326** may receive relevant data from the data receipt module **304**, which may flow through a scenario module **370** and a scenario efficacy module **374** to determine a relevant mitigation opportunity and an anticipated efficacy associated therewith. Specifically, such a scenario module **370** may be configured to generate one or more scenarios and scenario dataset **372** associated therewith.

[0058] For instance, such a scenario module may determine a first scenario in which no mitigations are made according to only that data relating to firm demand and firm supply of the primary supplier site **204**, such as EDP data **347**, sales order data **341**, pick list data **342**, bill of materials data **348**, sales forecast data **343**, inventory data **344**, purchase order data **345**, and work order data **346**. A second scenario may be determined according to the same information considered in the first scenario, except as applied from an expanded geographic perspective. A third scenario may instead append data relating to plan demand and plan supply—e.g., dependent demand data **349** and MRP data **350**—to generate a scenario in which external non-firm supply is used as a mitigation opportunity. A fourth scenario may additionally append the availability of safety stock onto the third scenario as yet another mitigation opportunity. As previously stated, each such generated scenario may carry with it a scenario dataset **372** detailing the specifics of the scenario, such as the number of supply needed for the mitigation opportunity, timeframes for delivery, etc. As may be understood, the foregoing scenarios are merely exemplary, and other mitigation opportunities borne out of scenarios generated by the scenario module **370** are envisioned herein. Likewise, one or more of said scenario datasets **372** may include the use of one or more secondary supplier sites **208***a*-**208***n*.

[0059] Using the scenario dataset(s) **372**, at least one embodiment of a scenario efficacy module **374** may be configured to employ one or more routines to determine whether a given scenario is anticipated to be effective in responding to a given disturbance event. For instance, a scenario supply routine **376** and a scenario demand routine **378** may be collectively employed to determine the degree to which supply and demand may be affected by the proposed scenario. Meanwhile a fixed BOM routine **380** may be utilized to collect bills of materials data **348** related to the scenario at issue from the data receipt module **304**. Collectively, the outputs from the scenario supply routine **376**, scenario demand routine **378**, and fixed BOM routine **380** may be fed into a pegging routine **382** configured to specifically link specific items of inventory data **344** with specific sales order data **341**, purchase order data **345**, and/or work order data **346** according to the supply, demand, and bill of materials data **348** generated from the foregoing routines. Subsequent thereto, a multi-level BOM routine **384** may be employed to generate a multi-level bill of materials. And, finally, a scenario status routine **386** may be used to determine an efficacy metric indicating the degree of efficacy a given scenario may have as a mitigation opportunity, whereas a readiness routine **388** may be employed to determine a deployment metric indicating whether the given scenario is ready for immediate deployment or some deployment in the future—e.g., this situation will be ready in three days when a given secondary supplier site **208***b* will have a surplus of inventory available after receiving a new shipment of raw materials. In at least one embodiment, such a scenario efficacy module **374** may additionally comprise an ID matching routine configured to identify and/or confirm items from inventory data **344** match within those contemplated in the given scenario. In at least one embodiment, such a scenario efficacy module **374** may generate a response output **398** in accordance with the foregoing.

[0060] Accordingly, the risk and recovery module **322**, inventory entitlement module **324**, and mitigation module **326** of at least one embodiment of the analytics system **310** may collectively predict the risk associated with a given disturbance event, proactively prioritize which sales order data **341** should be fulfilled in view of such a disturbance, organically calculate and adjust minimum inventory quantities through fluctuations caused by one or more disturbance events, and identify potential mitigation opportunities and the anticipated efficacy associated therewith. In so doing, at least one embodiment of the analytics system **310** may develop an end-to-end solution to a disturbance event, taking into consideration the various needs of each section of the supply chain and providing visibility to the various users thereof.

[0061] In at least one embodiment of the present disclosure, such as the ones depicted in FIGS. **3**A, **3**C, and **3**E, such an analytics system **310** may additionally comprise an exception module **390**. Such an exception module may be configured to automatically determine exception data **355** related to the disturbance event(s) generating constraint data **353** at a primary supplier site **204**. Such exception data **355** may comprise, for instance, a list of disturbance event(s) which may be ignored and/or deprioritized in relation to the response and decision-making processes of the supply chain platform **302**.

[0062] In at least one embodiment, such exception data **355** may be generated according to a consequence component **396** configured to determine appropriate exception data **355** in view of the data collected by the data receipt module **304**. For instance, such a consequence component **396** may be configured to assess the constraint data **353** in relation to historical data **351** associated disruption events previously responded to through the supply chain platform **302** described herein. For instance, such a consequence component **396** may utilize the constraint data **353** of the present disruption event to identify comparable constraint data **353** from a prior disruption event—i.e., a disruption event previously received and/or addressed by the supply chain platform **302**. Subsequent thereto, such consequence component **396** may compare response output(s) **398** associated with such prior disruption event to determine an exception likelihood—i.e., the likelihood the disruption event associated with the present constraint data **353** should be classified as an exception and thereby be identified as exception data **355**. As may be understood, such exception data **355** may be continuously adjusted as historical data **351** is generated by the supply chain platform **302**, wherein certain types of disruption events identified within the constraint data **353** may be added and/or removed from such exception data **355**.

[0063] In so doing, it may be understood the exception module **390** may be configured to differentiate between types of disruption events, such as by identifying important disruption events from insignificant disruption events. For instance, for one organization, a present disruption event —i.e., a disruption event presently at issue—at a primary supplier site **204** relating to a commonly produced article may be of little consequence—e.g., corrugated boxes may be common and manufactured at numerous secondary supplier sites **208***a*-**208***n*, and thus identifying an appropriate secondary supplier site **208***a* for remedial action may be considered an easy fix. In the same vein, such an exception module **390** may identify important disruption events such as those associated with rarer articles or those articles for which a delay will cause greater disruptions throughout the logistical structure **200**. And, similarly, such exception module **390** may identify, according to the historical data **351**, whether a disruption event previously identified as exception data **355** should be removed therefrom—e.g., although corrugated boxes are common, persistent issues at a primary supplier site **204** have exhausted overflow stock at the nearby secondary supplier sites **208***a*-**208***n*, and thus remediation of the present constraint data **353** will not be a routine matter.

[0064] In view thereof, the exception module **390** of various embodiments of the present disclosure may comprise one or more machine learning models configured to perform the aforementioned functions. Such machine learning models may include, without limitation, regression models, linear regression models, classification models, tree-based models, clustering models, or some combination thereof. As may be understood, the identification of the appropriate machine learning model(s) for the exception module **390** may, in certain embodiments, depend on the data collected by the data receipt module **304**, the historical data **351**, the logistical structure **200** of the organization, and, more generally, the structure of the supply chain system **100** itself. Hence, in at least one embodiment, such machine learning model(s) for use by the exception module **390** may be determined according to Monte Carlo simulations and/or other similar such methodologies.

[0065] In at least one embodiment, the exception data **355** generated by the consequence component **396** may be be validated by a user, such as through an exception list module **392** of the dashboard module **306**. Such an exception list module **392** may be configured to display the current exception data **355**, and relevant information associated therewith such as the relevant constraint data **353** associated with the determination thereof, to a user for validation. In so doing, a user may manually determine validation data **399** associated with the exception data **355**, which may be stored as historical data **351** for future use by the consequence component **396**. Alternatively, such validation data **399** may instead be determined according to the consequence component **396** itself, such as by referencing historical data **351** associated with exception data **355**, such as the results of actions taken, or the lack thereof, both before and after a disruption event was identified as exception data **355**.

[0066] With reference now to FIG. **3**E, it may be seen the exception module **390** may, in at least one embodiment, be utilized in connection with the mitigation module **326**. For instance, any data to be transmitted to the mitigation module **326** from the data receipt module **304** may first flow through such exception module **390**. In so doing, the exception module **390** may first filter out constraint data **353** for which no mitigation is imminently necessary. In so doing, the analytics system **310** may thus prioritize analyses for the most important disruption events.

[0067] Returning now to FIG. **3**A, it may be seen at least one embodiment of the supply chain platform **302** of the present invention may comprise a report module **312**. Such a report module **312** may be configured to generate one or more reports which may be displayed to a user operating a user device **114** through the UI component **320** of the dashboard module **306**. In at least one embodiment, such a report module **312** may be interactive such that the various report(s) generated thereby may be modified in real-time, such as through the key element component **316** and/or parameter component **318** of the dashboard module **306**. As may be understood, the various modules of which the report module **312** is comprised may each be configured to generate a different interactive report for a user. For instance, a glidepath module **328** may be configured to generate outputs relating to a glidepath report depicting risk and clearance projections, such as the material risk in view of a disruption event as well as the degree to which a predicted backorder may affect revenue expectation over a period of time.

[0068] A material risk module **330**, meanwhile may depict a variety of metrics relating to the material risk associated with a supply chain in the face of a disturbance event, such as the quantity of inventory at risk, the revenue loss associated therewith, and the degree to which production may be delayed over a period of time. In at least one embodiment, such a material risk module **330** may generate one or more interactive data visualization(s) **404**, such as an interactive heat map, depicting, for instance, the top mitigable risks. In so doing, such an interactive data visualization **404** may depict which products, goods, items, and/or materials are both at risk due to one or more disturbance events and remediable via one or more mitigation opportunities. However, it may be understood such an interactive data visualization **404**, whether comprising an interactive heat map or otherwise, may instead depict alternative portions of relevant data, such as the least mitigable risks.

[0069] In various embodiments of the present invention, such an interactive data visualization **404** may be configured for interaction by one or more users to enable a dynamic sensitivity analysis thereby. For instance, the parameter component **318** may be utilized to adjust the risk tolerance used by the analytics system **310**, adjust the allowable lapse period and/or allowable delay period, enable risk hedging, and/or enable varying degrees of risk exemptions. Accordingly, user(s) of the supply chain platform **302** may be enabled to adjust the sensitivity of the analytics system **310**, such as by specifying the type of risk to be considered, the type of mitigation opportunities falling within a prescribed timeframe, and the acceptable degree of risk tolerance for any primary supplier site **204**, secondary supplier site **208***a*-**208***n*, and/or distribution spoke **206***a*-**206***n*. As may be understood, such variety of adjustable parameterizations adjustable by the user(s) for such dynamic sensitivity analysis is merely exemplary, and alternative types of parameterizations and/or degrees thereof are envisioned herein. Likewise, it should be understood the foregoing dynamic sensitivity analysis may be applied to one or more reports and/or displays generated by the report module **312**, such that, in at least one embodiment of the present invention, the user selections afforded by the parameter component **318** and/or key element component **316** may permeate throughout the report module **312**.

[0070] Yet additional outputs of various embodiments of the report module **312** may comprise a shipping status module **332**, a quarterly status module **334**, a surplus status module **336**, an expiration status module **338**, and a portfolio module **340**. Such a shipping status module **332** may be configured to output data relating to, for instance, products that are built but yet to ship, along with commentary detailing any order lag associated with such shipment. Such a quarterly status module **334** may be configured to output data relating to the degree of concordance between an expected execution and an actual execution of the supply chain itself, or various portions thereof. The surplus status module **336** may be configured to output data relating to an surplus inventory, including a description of the inventory, an applicable stock keeping unit number, the inventory's location, and the amount of surplus, thereby enabling shorter lead times for customers and the ability to easily visualize potential business opportunities. Likewise, such an expiration status module **338** may be configured to output data relating to materials set to expire or otherwise having a do not sell after designation including, without limitation, data relating to the value of such materials, the timeframes for expiration, and fillable fields for user-input comments and/or designations. Such a portfolio module **340**, meanwhile, may generally output data relating to a portfolio of goods, products, materials, and/or inventory, including, for instance, a forward-looking view at risk and stock health as well as a support strategy for supply maintenance. As may be understood, such outputs of the various embodiments of the report module **312** are not exhaustive as additional outputs and accompanying modules are envisioned herein. Moreover, it may be understood the various modules and/or functions employed by the analytics system **310**, such as the risk and recovery module **322**, the inventory entitlement module **324**, and/or the mitigation module **326** may be applied and/or depicted within any such output of the report module **312**. Likewise, exception data **355** associated with the exception module **390** may be configured for application and/or display through the various embodiments of the report module.

[0071] With continued reference to FIG. **3**A, at least one embodiment of such a supply chain platform **302** may comprise a security module **314**. Such a security module **314** may be configured to provide secure transmission of data between the various systems and/or platforms interconnected with the supply chain platform **302**. Likewise, such a security module **314** may be configured to provide identity verification procedures for the user(s) operating the user device(s) **114**, such as suppliers, customers, and/or administrators.

[0072] FIGS. **4**A-**4**K generally depict various outputs of such report module **312** as generated, displayed, and/or provided to the user(s) through the UI component **320** of the dashboard module **306** for display on the user device(s) **114** and, particularly the display component **122** thereof. As may be understood, the user interface outputs depicted in such Figures is merely exemplary, as are the data, interactive data visualizations **404**, and other windows, buttons, fields, and other user interface elements depicted thereon.

[0073] With reference specifically to FIG. **4**A, it may be seen such UI component **320** is depicting at least one embodiment of an output of the glidepath module **328**, which may comprise a glidepath report **406**. As may be seen, such output may include an interactive data visualization **404** depicting, for instance, a line graph of a glidepath for a selected date range, and a bar chart for suppliers. Such output may further include a user input window **402** comprising a plurality of data fields, buttons, and other user interface elements through which a user may make selections, input comments, and indicate the status of a given piece of data relating to material risk.

[0074] FIGS. **4**B and **4**E likewise depict at least one embodiment of an output of a material risk module **330**, which may comprise a material risk report **408**. Such an output may include at least one interactive data visualization **404**, including an interactive heat map depicting top mitigable risks, as well as a variety of bar charts depicting various other portions of data. Likewise, as shown in FIG. **4**E, such an interactive data visualization **404** may, in at least one embodiment, additionally include a line graph having user selectable elements disposed thereon. In at least one embodiment, such user selectable elements may be configured to display additional data upon selection thereof. Such an output may, in at least one embodiment, further include a user interface representation of a parameter component **318**, through which a user may selectively apply parameter data **354**, as discussed in greater detail heretofore. Such embodiments of the user interface representation of the parameter component **318** and an interactive data visualization **404** are depicted in FIGS. **4**C and **4**D for further reference.

[0075] Depicted in FIGS. **4**E-**4**I are various embodiments of the outputs of the shipping status module **332**, quarterly status module **334**, surplus status module **336**, and expiration status module **338**, respectively. FIGS. **4**J and **4**K, meanwhile, each depict at least embodiment of the portfolio module **340**. As may be noted, each such output—e.g., a shipping status report **410**, a quarterly status report **412**, a surplus status report **414**, an expiration status report **416**, and a portfolio report **418**—may include an interactive data visualization **404** and/or a user input window **402**, each of which may function in accordance with the disclosure recited heretofore. As may be noted, each of such outputs may depict different portions of data, any one of which may stem from the various portions of the supply chain platform **302**.

[0076] In view thereof, various embodiments of the supply chain system **100** disclosed herein are configured to increase the efficiency and adaptability of supply chains operated by vertically integrated organizations. Such supply chain system **100** may provide greater visibility across an organization, provide predictive analyses, and develop prescriptive plans in response to a disruptive event. Furthermore, such supply chain system **100** may provide insights between various segments of the organization and harmonize designs and strategies to facilitate site-to-site sharing. Such supply chain system **100** may further be user-configurable to enable specific parameterization of the analyses of outputs thereof. And, in various embodiments, such a supply chain system **100** may be configured to identify and differentiate between important and inconsequential disruptive events at primary supplier site(s) **204** to generate exception data **355**, thereby enabling the supply chain platform **302** to assess, remediate, and display information most useful to a user.

Claims

1. A supply chain system comprising: an inventory management hub communicatively configured in connection with a supply chain platform; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub; an analytics system configured to determine at least one risk score according to a risk calculation routine performed on said supply chain data; said analytics system further configured to generate at least one mitigation opportunity according to a mitigation module; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity.

2. The supply chain management system of claim 1, wherein said supply chain data comprises sales order data, pick list data, sales forecast data, inventory data, purchase order data, work order data, EDP data, bill of materials data, dependent demand data, and MRP data.

3. The supply chain management system of claim 1, wherein said supply chain platform further comprises a historical data module configured to generate historical data.

4. The supply chain management system of claim 1, wherein said supply chain platform further comprises a dashboard module.

5. The supply chain management system of claim 4, wherein said dashboard module comprises a key element component configured to receive key element data from said at least one user device.

6. The supply chain management system of claim 4, wherein said dashboard module comprises a parameter component configured to receive parameter data from said at least one user device, said parameter data configured to apply at least one parameterization to said at least one interactive data visualization.

7. The supply chain management system of claim 1, wherein said analytics system is further configured to generate exception data through an exception module, said exception data determined according to said historical data.

8. The supply chain management system of claim 7, wherein said exception module comprises at least one machine learning model configured to determine said exception data by comparing first constraint data for a present disruption event at a primary supplier site with said historical data, said historical data comprising second constraint data and at least one response output from a prior disruption event, said at least one response output generated by said analytics system in relation to said prior disruption event.

9. A supply chain management system comprising: an inventory management hub communicatively configured in connection with a supply chain platform; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub; a historical data module configured to generate historical data; an analytics system comprising a risk and recovery module configured to determine at least one risk score according to a risk calculation routine performed on said supply chain data and said historical data; said analytics system further comprising a mitigation module configured to generate at least one mitigation opportunity according to said supply chain data and said historical data; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity.

10. The supply chain management system of claim 1, wherein said supply chain platform further comprises a dashboard module, said dashboard module comprising: a key element component configured to receive key element data from said at least one user device; and a parameter component configured to receive at least one parameter input from said at least one user device.

11. The supply chain management system of claim 10, wherein said risk and recovery module is further configured to generate at least one risk classification according to at least said key element data.

12. The supply chain management system of claim 10, wherein said at least one interactive data visualization is configured for parameterization according to said at least one parameter input.

13. The supply chain management system of claim 9, wherein said mitigation module comprises a scenario module configured to generate at least one scenario dataset.

14. The supply chain management system of claim 13, wherein said mitigation module further comprises a scenario efficacy module configured to determine to at least one efficacy metric and at least one deployment metric for said at least one scenario dataset.

15. A supply chain management system comprising: an inventory management hub communicatively configured in connection with a supply chain platform and a supplier system for a logistical structure, said logistical structure comprising a primary supplier site and at least one secondary supplier site, said primary supplier site configured to generate constraint data in relation to a disruption event; said supply chain platform communicatively configured in connection with at least one customer system and at least one user device; said supply chain platform comprising a supply chain server system configured to perform at least one process of said supply chain platform; said supply chain platform further comprising: a data receipt module configured to receive said supply chain data from said inventory management hub and said constraint data from said primary supplier site; a historical data module configured to generate historical data; a dashboard module comprising a key element component configured to receive key element data from said at least one user device and a parameter component configured to receive parameter data from said at least one user device; an analytics system comprising a risk and recovery module configured to determine, in response to said constraint data, at least one risk score and at least one risk classification according to a risk calculation routine performed on said supply chain data and said historical data; said analytics system further comprising a mitigation module configured to generate at least one mitigation opportunity, said at least one mitigation opportunity generated according to a scenario module configured to determine at least one scenario dataset and a scenario efficacy module configured to determine an efficacy metric and a deployment metric for said at least one scenario dataset; and a report module configured to generate at least one interactive data visualization for display on said at least one user device, said at least one interactive data visualization based on said at least one risk score and said at least one mitigation opportunity, and configurable according to said at least one parameter input.

16. The supply chain management system of claim 15, wherein said at least one mitigation opportunity includes the use of said at least one secondary supplier site.

17. The supply chain management system of claim 15, wherein said dashboard module further comprises a UI component configured to generate at least one user interface according to at least one output of said report module for display on said at least one user device.

18. The supply chain management system of claim 17, wherein said report module comprises: a glidepath module configured to generate a glidepath report; a material risk module configured to generate a material risk report; a shipping status module configured to generate a shipping status report; a quarterly status module configured to generate a quarterly status report; a surplus status module configured to generate a surplus status report; an expiration status module configured to generate an expiration status report; and a portfolio module configured to generate a portfolio report.

19. The supply chain management system of claim 15, wherein said risk and recovery module further comprises a demand prioritization routine configured to determine a demand priority metric.

20. The supply chain management system of claim 15, wherein said analytics system further comprises an inventory entitlement module configured to determine an inventory classification score and a service level score according to an inventory quantity routine.
